

DELCODE-PET | Florbetaben: Semi-quantitative SUVR and Centiloid values

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Purpose:

This document provides background information for the semiquantitative standard uptake value ratio (SUVR) analyses that were performed for Neuraceq™ (18F-Florbetaben [FBB]: Life Radiopharma, Berlin) amyloid PET scans in the DELCODE-PET study.

The original SUVR analyses of DELCODE-PET FBB scans were subject to a research agreement with Life Molecular Imaging. Moreover, the final FBB examinations were performed with support of the AMYPAD consortium. Therefore, article or poster acknowledgments which use this information should contain the information: *“Initial data analyses of the Florbetaben PET exams reported in this study were subject of a research agreement with Life Molecular Imaging. Some Florbetaben PET examinations were performed with support of the AMYPAD consortium”*.

Analyzed FBB data:

In general, FBB PET examinations were **only collected from Subjective Cognitive Decline (SCD) participants from the DELCODE study** who agreed to these additional examinations.

No complementary PET data are available from healthy controls, MCI and AD dementia participants!

FBB PET examinations were **performed up to two times**: A baseline scan (“FBB_P0”) and a follow-up scan (“FBB_P24”).

Note: Due to the delayed start of DELCODE-PET, the baseline scan was not always performed at the baseline visit of the DELCODE main study, but during follow-up visits of the DELCODE main study. Accordingly, the “P0” and “P24” labels for the PET data are not equivalent with the “M0, M12, M24, ...” labels found in the MRI data. Moreover, while follow-up PET scans were originally scheduled for a 24-month delay (hence: “P24”), this was later extended to a time window of up to 36 months.

FBB data acquisition:

PET data acquisition followed established standard protocols.

	FBB
Applied tracer activity	260-300 Megabequerel (MBq)
Start of emission scan	90 min post-injection
Duration of emission	20 min
Acquisition mode	3D Listmode, <i>post hoc</i> reconstruction in 4*5 min time frames
Reconstruction algorithm	<i>Iterative:</i> Based on established brain protocols at PET study sites • <i>Siemens + GE:</i> Ordered Subset Expectation Maximization (OSEM) • <i>Philips:</i> Line-of-response row-action maximum likelihood (LOR-RAMLA)
Image corrections	• Decay • Randoms • Dead time • Scatter • Attenuation • Normalization
Attenuation correction	Based on vendor-specific attenuation coefficient (μ) maps • <i>Siemens/ GE/ Philips PET/CT systems:</i> Derived from low-dose CT • <i>Siemens PET/MR systems:</i> Derived from DIXON-VIBE MR sequence

Data analyses:

Standardized Uptake Value Ratios (SUVR) were calculated as a semi-quantitative measure of FBB tracer uptake. These values serve as proxy for cortical gray matter amyloid depositions in the brain, relative to a reference region (here: cerebellar cortex) which is assumed to show low tracer uptake.

Most studies use **global SUVR** values, i.e. aggregate tracer uptake values over a composite of cortical brain regions (for an excellent state-of-the-art review of methods: [Pemberton et al, 2022](#)).

We provide two alternative forms of global SUVR calculation. “**Barthel SUVR**” will refer to SUVR calculations that used similar aggregated cortical volumes of interest (VOI) and cerebellar cortical reference regions like the “classical” validation studies for the FBB tracer, which gave us the opportunity to adopt the respective cut-off values for amyloid positivity ([Barthel et al., 2011](#); [Sabri et al., 2015](#)). While we still provide these calculations in the established form for continuity purposes, most users will probably prefer the newly introduced **Centiloid** (CL) score calculations.

“Classical” SUVR (Barthel)

Data analysis was **performed in PMOD** (PMOD Technologies LLC, Zurich, Switzerland), using the *Maximum Probability Atlas* processing pipeline implemented in PNEURO (Figure 1), analogous to the Florbetaben reference paper ([Barthel et al., 2011](#)).

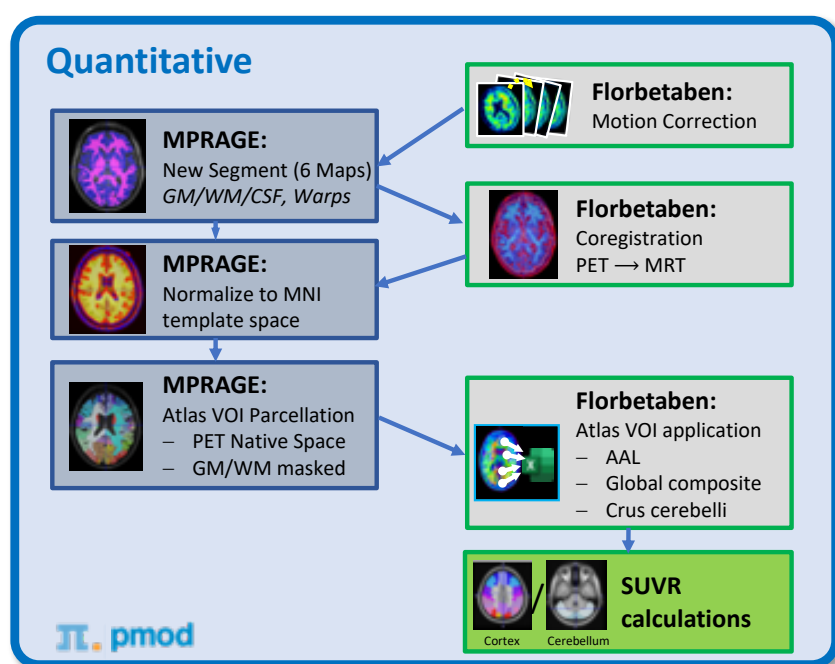


Figure 1: Florbetaben SUVR calculations with the PNEURO Maximum Probability Atlas pipeline

Linear transformations were used for motion correction of the FBB image frames, and coregistration to the MPRAGE anatomical MRI image of the respective patient. The MPRAGE scan was segmented (using a routine with 6 tissue classes similar to “New Segment” in SPM12) to (a) derive normalization parameters (which allow non-linear warping of imaging data between patient and atlas template space), and (b) calculate individual tissue probability maps for gray matter (GM), white matter (WM) and cerebrospinal fluid (CSF).

These spatial transformations were used to warp the AAL atlas template ([Tzourio-Mazoyer et al., 2002](#)) into the *native* PET space of the individual, and derive anatomical volumes of interest (VOI) definitions which were additionally masked inclusively with the GM probability map (default threshold: 50%), and exclusively with the CSF probability map (default threshold: 50%). To date, the processing pipeline does not yet include partial volume effect corrections.

For each of the delineated VOI, the average voxel value was calculated and extracted for further data processing. To examine cortical tracer uptake, a volume-weighted average of

- frontal,
- lateral temporal,
- parietal,
- occipital,
- anterior cingulate and
- posterior cingulate

anatomical VOI was calculated. The value for this global composite VOI was scaled by the value of the reference VOI (defined by the crus cerebelli of both hemispheres), to derive the main outcome variable, “global FBB SUVR”.

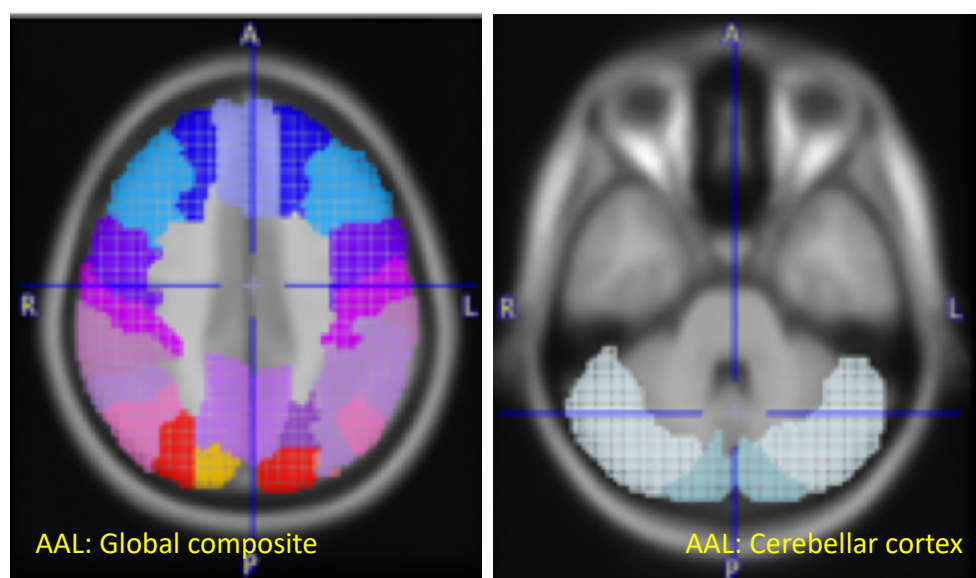


Figure 2: Anatomical regions in the AAL template used for delineation of global composite and cerebellar reference VOI. Note that these actual volumes of interest were smaller, because they were additionally masked with the gray matter probability maps ($p > .5$) of the individual participant.

The selection of the anatomical regions was based on the original paper (H. Barthel, personal communication) to preserve compatibility with previous publications ([Barthel et al., 2011](#); [Sabri et al., 2015](#)).

Note: The average activity uptake values for each single GM region of the AAL atlas that are extracted from PMOD for *post hoc* aggregation of volume-weighted composite VOI are currently *not* prepared for general release because a systematic empirical evaluation of these single region values is lacking. Meanwhile, there are technical options to generate tailored VOI composites for qualified research questions. In this case, please contact the PET group for further discussion.

Centiloid

Because there are various tracers with slightly different characteristics, and various analytical strategies (software packages, brain atlases, target and reference region choices, etc.), there is growing interest to standardize amyloid load quantification and make observations from different tracers and/or study cohorts more comparable. The Centiloid protocol ([Klunk et al., 2015](#)) is an initial step to achieve this goal. It did not only provide (1) a processing framework that – if performed appropriately – allows to produce Pittsburgh Compound B (PiB) SUVR calculations technically similar to the original study. Because the original study examined both young participants (=probably amyloid-negative) and clinical AD patients (=probably amyloid-positive), they could also (2) use the observed mean PiB SUVR of these two groups to define “0” (=typical amyloid-negative) vs “100” (=typical amyloid-positive) reference points on a common “Centiloid” scale.

Note: This means that amyloid-negative individuals with SUVR levels below the young controls’ *average* can have $CL < 0$, while amyloid-positive individuals with SUVR above the AD patients’ *average* can have $CL > 100$.

While Centiloid was originally defined with PiB data (“Level-1 analysis” in [Klunk et al., 2015](#)), later studies collected PET data with both PiB and alternative amyloid tracers (including FBB: [Rowe et al., 2017](#)) *in the very same individuals*, and found that the SUVR values derived with the Centiloid pipeline were numerically different, but highly correlated between PiB and the alternative tracers. If sufficiently high, this allows regression-based linear transformations of the tracer SUVR to PiB SUVR *equivalents*, which can subsequently be transformed into Centiloid values (“Level-2 analysis” in [Klunk et al., 2015](#)). This way, an (approximately) common scaling of the SUVR values across different tracers becomes possible. Importantly, these pipelines can be adopted by other sites (“Level-3 analysis” in [Klunk et al., 2015](#)), if the given implementation of the analysis pipeline (e.g. in SPM, PMOD) does not introduce error. This is empirically tested by reanalyzing the original datasets, and comparing the results with the results from the original paper.

Data analysis was again **performed in PMOD**, using the PNEURO *Maximum Probability Atlas* processing pipeline with a modified version of the Centiloid atlas provided in the software. The processing steps and parameters followed the detailed description provided in an [Application Note by PMOD](#).

The main processing steps include:

- Preparatory motion correction and averaging of the original 4 PET time frames
- Automated cropping of the MRI and PET images
- Segmentation of the MRI image (using the SPM8 Unified Segmentation approach with three tissue classes for GM/WM/CSF)
- Coregistration of the PET image to the MRI image
- Normalization of the MRI and PET images (i.e. warping into MNI space)
- Delineation of the Centiloid atlas VOI for the cortical target region (CTX) and the whole cerebellum (WC) reference region in atlas (not: native) space
- Extraction of mean voxel values for CTX and WC
- Calculation of the ratio: $SUVR = CTX/WC$
- Initially, the SUVR were transformed to Centiloid simply by using the original formula from [Rowe et al., 2017](#): $CL = 153.4 * TracerSUVR - 154.9$. *This was updated*, replacing the original parameters with a pipeline-specific formula (explained below): $CL = 154.472 * TracerSUVR - 155.661$.

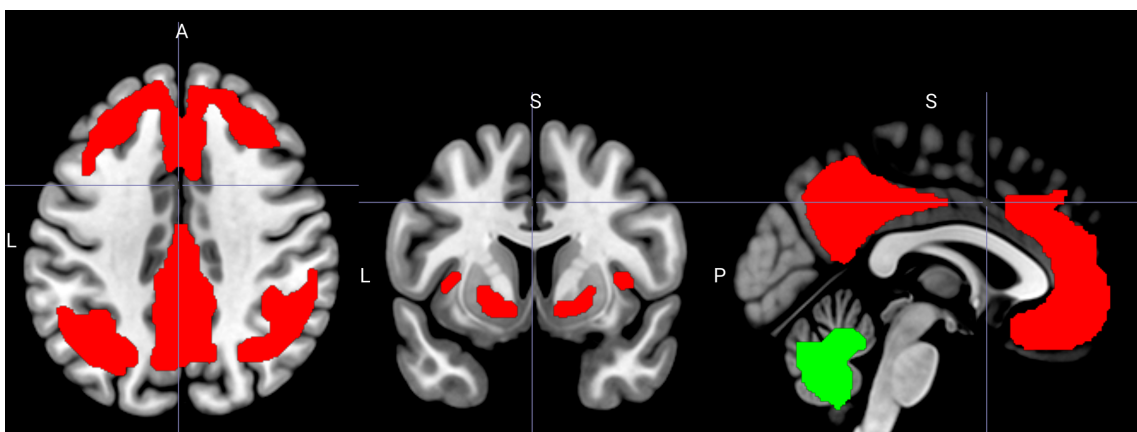


Figure 3: Cortical target (CTX) and Whole Cerebellum reference region (WC) used for Centiloid analysis, projected onto the MNI-152 brain template in MRICroGL¹.

Important technical information: The applicability of the linear transformation formulae above depends on the assumption that data processing reproduces the original Centiloid processing pipeline with sufficient accuracy, especially since we did not use the original SPM8 software package, but an independent software.

¹ Note that the Centiloid target region is more restricted than the cortical region used in the Barthel SUVR calculation. Moreover, there is no gray matter masking of the VOI regions, and the reference region includes both gray and white matter of the cerebellum. Despite these technical differences, preliminary analyses of DELCODE baseline datasets suggest that the correlation between the two measures is quite high ($r=0.966$).

To assure this, the Centiloid workgroup *demands* empirical validation, which is based on a mandatory re-analysis of both the original PiB ([Klunk et al. \(2015\)](#)) and FBB datasets (**Fehler! Linkreferenz ungültig.**), freely available from www.gaain.org/centiloid-project. The calculated SUVR and CL values have to be statistically compared with the original data, and users may be asked to provide this benchmark information somewhere (e.g. in a supplement):

- *PiB ([Klunk et al., 2015](#)): PMOD mean $SUVR_{CTX/WC}$ values were $M(AD)=2.074$ and $M(YC)=1.011$ for AD patients and young controls, respectively, less than 2% different from the original results. Using these sample parameters for Centiloid scaling, regression of the PMOD-calculated Centiloid values with the original CL values fulfilled the established quality criteria (slope=0.999, intercept=-0.04, $R^2=0.999$).*
- *FBB ([Rowe et al., 2017](#)): PMOD mean $SUVR_{CTX/WC}$ values were $M(elderly)=1.43$ and $M(young)=1.01$ for elderly and young participants, respectively, less than 2% different from the original results. Regression of PMOD-calculated Centiloid values with the original CL values fulfilled the established quality criteria (slope=1.005, intercept=-1.103, $R^2=0.999$). For direct transformation of PMOD-calculated FBB SUVR into PMOD-calculated CL, linear regression provided the following equation:

$$^{FBB}CL = 154.472 * ^{FBB}SUVR - 155.661.$$*

In general, it should be pointed out that the Centiloid approach only allows to achieve *approximate* comparability across tracers and study cohorts: For example, the original authors ([Klunk et al., 2015](#); [Rowe et al., 2017](#)) note that potential technical influences of different PET camera systems, reconstruction protocols, etc. have not been investigated systematically yet (which is still valid). This may contribute to the observation that pooled data analyses (e.g. [Salvado et al., 2019](#)) make converging observations, but still find some inter-sample variation, e.g. regarding cut-off values for amyloid positivity (e.g. [Farrell et al., 2021](#)).

To reduce the potential effects of different scanner resolutions, we performed two Centiloid analysis for each DELCODE-PET dataset: One using the original PET image series, and one using a recently finalized preprocessed version of the PET datasets with harmonized voxel geometry image resolution of ~8 mm full-width-half-maximum (FWHM) across PET scanners (similar to what is done in ADNI: [Jagust et al., 2015](#)).

General limitations:

- No quantifications are available for participants without usable MPRAGE scan, and for some cases without recent MR scans (primarily relevant for follow-up measurements), only preliminary SUVR and CL calculations based on older MR data can be provided that should be used with caution. At least for the Centiloid pipeline, the broad outline of the CTX region in normalized space (Figure 3) makes it likely that it is robust against smaller atrophic changes. In the future, it may be possible to solve this issue by using PET-only pipelines: Recent studies suggest strong convergence with the MR-based Centiloid pipeline (e.g. [Iaccarino et al., 2022](#)).
- No quantifications are available for participants where morphological abnormalities (cysts, meningioma etc.) precluded reliable delineation of regional VOI definitions.
- For some participants, the injected tracer dose was lower than defined in the DELCODE protocol (260-300 Megabequerel [MBq]), but still consistent with the manufacturer's suggestions (300±60 MBq) in most cases. To allow a more stringent data selection, SUVR/CL values are provided along with the injected tracer dose.
- For some participants, emission scan did not start 90 minutes post-injection [p.i.], usually due to technical problems. There is preliminary empirical evidence ([Barthel et al., Journal of Nuclear Medicine 2017](#)) that the Florbetaben SUVR values are robust across various time windows between 45 and 130 minutes. To allow a more stringent selection, the global SUVR values are provided along with the start time of the emission scan (in min p.i.).

Available variables:

The main variables are the semiquantitative parameters for global cortical tracer binding:

FBB_SUVR_P0 FBB_SUVR_P24	<i>FBB: Standard Uptake Value Ratio</i> Following Barthel et al. (2011): Relative to cerebellar crus
FBB_CL_P0 FBB_CL_P24	<i>FBB: Centiloid (original)</i> Based on the original image series
FBB_CL_8mm_P0 FBB_CL_8mm_P24	<i>FBB: Centiloid (8mm resolution)</i> Based on preprocessed FBB PET images with harmonized image resolution of 8 mm FWHM

Additionally, various background variables are provided:

FBB_SUVR_COMM_P0 FBB_SUVR_COMM_P24	<i>Quality comments</i> Mainly regarding limitations to usability, especially lack of recent MR data. Applies to both Barthel SUVR and Centiloid calculations.
FBB_DATE_P0 FBB_DATE_P24	<i>Date of PET examination</i>
FBB_REF_P0 FBB_REF_P24	<i>Clinical reference visit</i> Especially for FBB, PET exam dates could fall in-between two consecutive DELCODE main study visits. Here, the study visit <i>closer in time</i> is suggested for reference (which can be the next one).
AGE_FBB_P0 AGE_FBB_P24	<i>Age at time of PET examination</i>
FBB_DOSE_P0 FBB_DOSE_P24	<i>Decay-corrected injected tracer activity [in Megabequerel, MBq]</i> For some participants, the injected FBB tracer dose was lower than defined in the DELCODE protocol (260-300 MBq), but still consistent with manufacturer's suggestions (300±60 MBq) in most cases. This variable can be used for selecting participants more stringently, if needed.
FBB_DELAY_P0 FBB_DELAY_P24	<i>Time between tracer injection and start of PET emission [in min]</i> Based on available DICOM and checklist information. For some participants, emission scan did not start regularly, i.e. 90 min (FBB) post-injection, usually due to technical problems. Users can use this variable to select cases more stringently.

FBB_PETCAM_P0 FBB_PETCAM_P24	<i>Vendor and model of the PET scanner</i> • GE Discovery 690: PET/CT • Philips Gemini TF 16 TOF: PET/CT (@two sites) • Siemens Biograph Duo: PET/CT • Siemens Biograph mCT 128 Flow Edge: PET/CT • Siemens Biograph mMR: PET/MRI (@two sites)
FBB_DEVICE_P0 FBB_DEVICE_P24	<i>PET scanner ID</i> For optional use of scanner identity as a statistical covariate. Note that some clinical sites used the same PET scanner. Identical with <i>Device Serial Number</i> (0018,1000) in DICOM header. Because the original vendor information was already deleted by some PET sites before upload, and for pseudonymization, each single scanner received a new, arbitrary ID string ("00001"- "00007").
FBB_RECON_P0 FBB_RECON_P24	<i>PET reconstruction algorithm</i>
FBB_GEO_P0 FBB_GEO_P24	<i>Original PET image geometry</i> Reconstruction matrix, voxel size, slice number, etc.

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